Recap

Sometimes we care about predictions ***after our intervention***

We want to ask " ***What if…?*** " type of questions

This requires us to know what ***drives (causes)*** the changes in the dependent variable

Causal relationship satisfies three criteria (by John Stuart Mill):
1) The "cause" precedes the "effect" in time (temporal precedence),
2) The "cause" and the "effect" are related (covariation), and
3) There are no plausible alternative explanations for the observed covariation (non-spuriousness).



Randomized Experiment

Recap

We cannot directly observe an individual's causal effect.

If one could rewind history, and change only the action, then causation could be observed.

"No man ever steps in the same river twice, for it's not the same river and he's not the same man."

Potential Outcome Framework

Definition (Causal Effect, or Unit Treatment Effect)

Causal effect of the treatment on the outcome for unit i is the difference between its two potential outcomes: $\tau_i \triangleq Y_i(1) - Y_i(0)$

Stable Unit Treatment Value Assumption (SUTVA)

Assume there are N people in the population, then the ***average causal effect*** of the population is

$$\tau_{fs} = \frac{1}{N}\sum_{i=1}^{N}[Y_i(1) - Y_i(0)] = \bar{Y}(1) - \bar{Y}(0)$$

Recap

Goal: estimate the population average causal effect using ***observed outcomes***

How those observed outcomes are generated matters. (Assignment Mechanism)

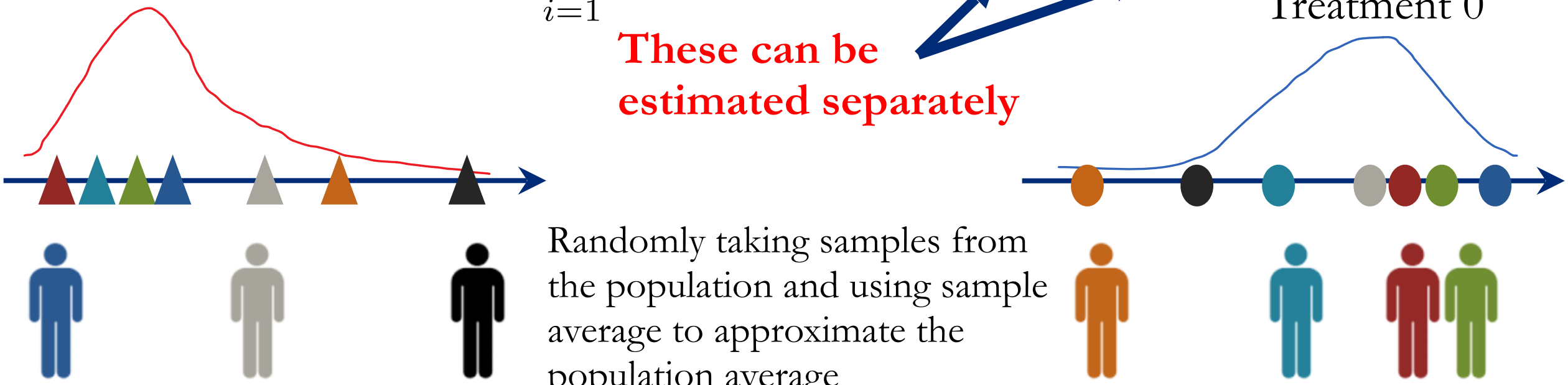
Conceptually, imagining that you make a copy of the experiment population

$$\tau_{fs} = \frac{1}{N}\sum_{i=1}^{N}[Y_i(1) - Y_i(0)] = \bar{Y}(1) - \bar{Y}(0)$$

Treatment 1

Treatment 0

These can be estimated separately



Randomly taking samples from the population and using sample average to approximate the population average

Recap

You cannot first check each individual's $(Y_i(1), Y_i(0))$ and then decide whether to sample this one from the Y_1 population or Y_0 population. Because these samples won't be ***representatives*** of their population.

$$\tau_{fs} = \frac{1}{N}\sum_{i=1}^{N}[Y_i(1) - Y_i(0)] = \bar{Y}(1) - \bar{Y}(0)$$

Treatment 1

Treatment 0

These can be estimated separately

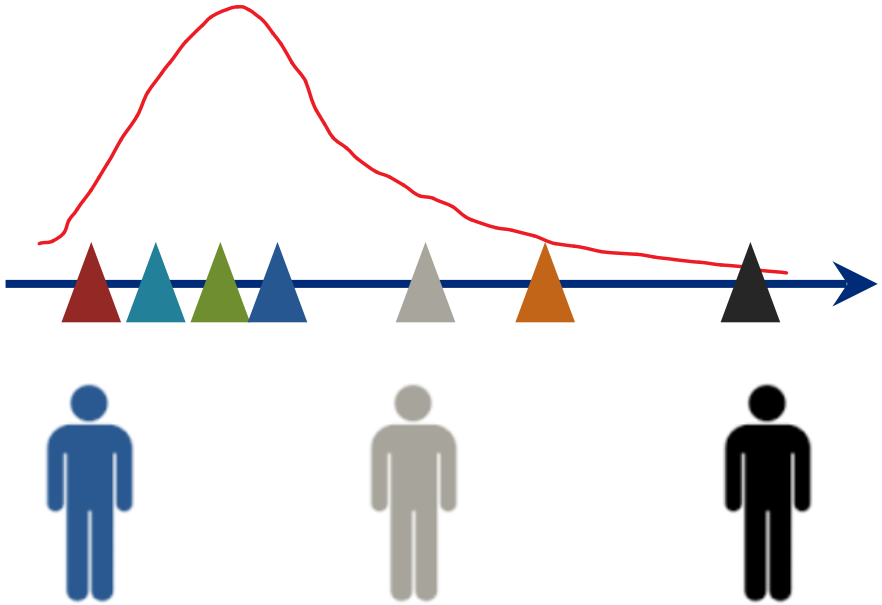


"Sampling" should be independent of each individual's potential outcome and also ensures the same subject cannot be sampled from both groups.

Randomized Assignment

Goal: estimate the population average causal effect using ***observed outcomes***

Treatment 1



For each subject in the experiment, flip a coin to determine the assignment, e.g., Head to treatment 1 and Tail to treatment 0.

Treatment 0



$$\hat{\tau}_{fs} = \bar{Y}_T^{obs} - \bar{Y}_C^{obs}$$

$$\approx$$

$$\tau_{fs} = \bar{Y}(1) - \bar{Y}(0)$$

$$\bar{Y}_T^{obs} = \frac{1}{N_T} \sum_{i \in T} Y_i(1)$$

$$\bar{Y}_C^{obs} = \frac{1}{N_C} \sum_{i \in C} Y_i(0)$$

The condition for "$\approx$" is: the assignment is independent of the potential outcome

Fisher's Exact Test

With randomized assignment, the resulting estimator $\hat{\tau}_{fs} = \bar{Y}_T^{obs} - \bar{Y}_C^{obs}$ is an unbiased estimator of the population average causal effect $\tau_{fs} = \bar{Y}(1) - \bar{Y}(0)$

$$\mathbb{E}[\hat{\tau}_{fs}] = \tau_{fs}$$

τ_{fs} is a constant

$\hat{\tau}_{fs}$ is a random variable, as you randomly assigned some subjects to the treatment group and observe their potential outcome under the treatment. If you repeat the randomized assignment procedure again, you will like to have a different set of subject being assigned to the treatment group.

In the experiment, you only perform randomized assignment once. Then how can you interpret the value $\hat{\tau}_{fs} = \bar{Y}_T^{obs} - \bar{Y}_C^{obs}$ from your experiment?

Fisher's Exact Test

In 1935, Fisher was intent on assessing the *sharp null hypothesis* of no effect of the active versus control treatment

H_0: for each unit in the experiment, both values of the potential outcomes are identical

$$H_0 : Y_i(1) = Y_i(0), \ \forall \ i$$

But $Y_i(1)$ can still be different from $Y_j(1)$ for $i \neq j$

Fisher developed methods for calculating "p-value" for this specific hypothesis, which we refer to as *Fisher Exact P-values*

$$\text{General } H_0 : Y_i(1) = f\left(Y_i(0)\right), \ \forall \ i.$$

f is known and one-to-one

Fisher's Exact Test

We randomly select 3 of the 7 subjects for treatment; the remaining 4 form the control group. Every set of 3 subjects is equally likely to be selected. The dependent variable is Y

Subject ID	Potential Outcome			Assignment	Observed Outcome
	$Y_i(0)$	$Y_i(1)$		D_i	Y_i^{obs}
1	(?)	5		1	5
2	4	(?)		0	4
3	6	(?)		0	6
4	(?)	7		1	7
5	3	(?)		0	3
6	1	(?)		0	1
7	(?)	3		1	3

Fisher's Exact Test

Fisher Sharp Null Hypothesis $H_0 : Y_i(1) = 2 \cdot Y_i(0), \ \forall \ i$

Or equivalently $H_0 : Y_i(1)/Y_i(0) = 2, \ \forall \ i$

Subject ID	Potential Outcome			Assignment	Observed Outcome
	$Y_i(0)$	$Y_i(1)$		D_i	Y_i^{obs}
1	**(2.5)**	5		1	5
2	4	**(8)**		0	4
3	6	**(12)**		0	6
4	**(3.5)**	7		1	7
5	3	**(6)**		0	3
6	1	**(2)**		0	1
7	**(1.5)**	3		1	3

Fisher's Exact Test

Pick a statistic $T(\mathbf{D}, \mathbf{Y}^{obs}) = \bar{Y}^{obs}(1)/\bar{Y}^{obs}(0)$

The randomness of the statistic comes from the randomized assignment

Subject ID	Potential Outcome			Assignment	Observed Outcome
	$Y_i(0)$	$Y_i(1)$		D_i	Y_i^{obs}
1	**(2.5)**	5		1	5
2	4	**(8)**		0	4
3	6	**(12)**		0	6
4	**(3.5)**	7		1	7
5	3	**(6)**		0	3
6	1	**(2)**		0	1
7	**(1.5)**	3		1	3

Fisher's Exact Test

Compute the value of your statistic $T(\mathbf{D}, \mathbf{Y}^{obs}) = \bar{Y}^{obs}(1)/\bar{Y}^{obs}(0)$

What is the value of $T(\mathbf{D}, \mathbf{Y}^{obs}) = \bar{Y}^{obs}(1)/\bar{Y}^{obs}(0)$ here?

$\bar{Y}^{obs}(1)/\bar{Y}^{obs}(0) = [(5+7+3)/3]/[(4+6+3+1)/4] = 1.4286$

Subject ID	Potential Outcome			Assignment	Observed Outcome
	$Y_i(0)$	$Y_i(1)$		D_i	Y_i^{obs}
1	**(2.5)**	5		1	5
2	4	**(8)**		0	4
3	6	**(12)**		0	6
4	**(3.5)**	7		1	7
5	3	**(6)**		0	3
6	1	**(2)**		0	1
7	**(1.5)**	3		1	3

Fisher's Exact Test

$$\bar{Y}^{obs}(1)/\bar{Y}^{obs}(0) = [(5+7+3)/3]/[(4+6+3+1)/4] = 1.4286$$

If the assignment looks like this, then what is value of $\bar{Y}^{obs}(1)/\bar{Y}^{obs}(0)$?

$$\bar{Y}^{obs}(1)/\bar{Y}^{obs}(0) = [(8+6+2)/3]/[(2.5+6+3.5+1.5)/4] = 1.5802$$

Subject ID	Potential Outcome			Assignment	Observed Outcome
	$Y_i(0)$	$Y_i(1)$		D_i	Y_i^{obs}
1	2.5	5		0	2.5
2	4	8		1	8
3	6	12		0	6
4	3.5	7		0	3.5
5	3	6		1	6
6	1	2		1	2
7	1.5	3		0	1.5

Hypothetical assignment and outcomes under H_0

Fisher's Exact Test

Derive the distribution of your statistic $T(\mathbf{D}, \mathbf{Y}^{obs}) = \bar{Y}^{obs}(1)/\bar{Y}^{obs}(0)$ under H_0 and the randomized assignment mechanism

The randomized assignment mechanism is specified by the experimenter. Hence, the experimenter knows $\mathbb{P}\{\mathbf{D} = \mathbf{d}\}$

Subject ID	Potential Outcome			Assignment	Observed Outcome
	$Y_i(0)$	$Y_i(1)$		D_i	Y_i^{obs}
1	(2.5)	5		1	5
2	4	(8)		0	4
3	6	(12)		0	6
4	(3.5)	7		1	7
5	3	(6)		0	3
6	1	(2)		0	1
7	(1.5)	3		1	3

$\mathbf{d} = (1, 0, 0, 1, 0, 0, 1)$

Fisher's Exact Test

In this example, the randomization procedure we used is: randomly assign 3 subject (out of 7 subject) to the treatment group and the remaining to the control group.

How many possibilities we can have? 7 choose 3 = 35

Subject ID	Potential Outcome			Assignment	Observed Outcome
	$Y_i(0)$	$Y_i(1)$		D_i	Y_i^{obs}
1	**(2.5)**	5		1	5
2	4	**(8)**		0	4
3	6	**(12)**		0	6
4	**(3.5)**	7		1	7
5	3	**(6)**		0	3
6	1	**(2)**		0	1
7	**(1.5)**	3		1	3

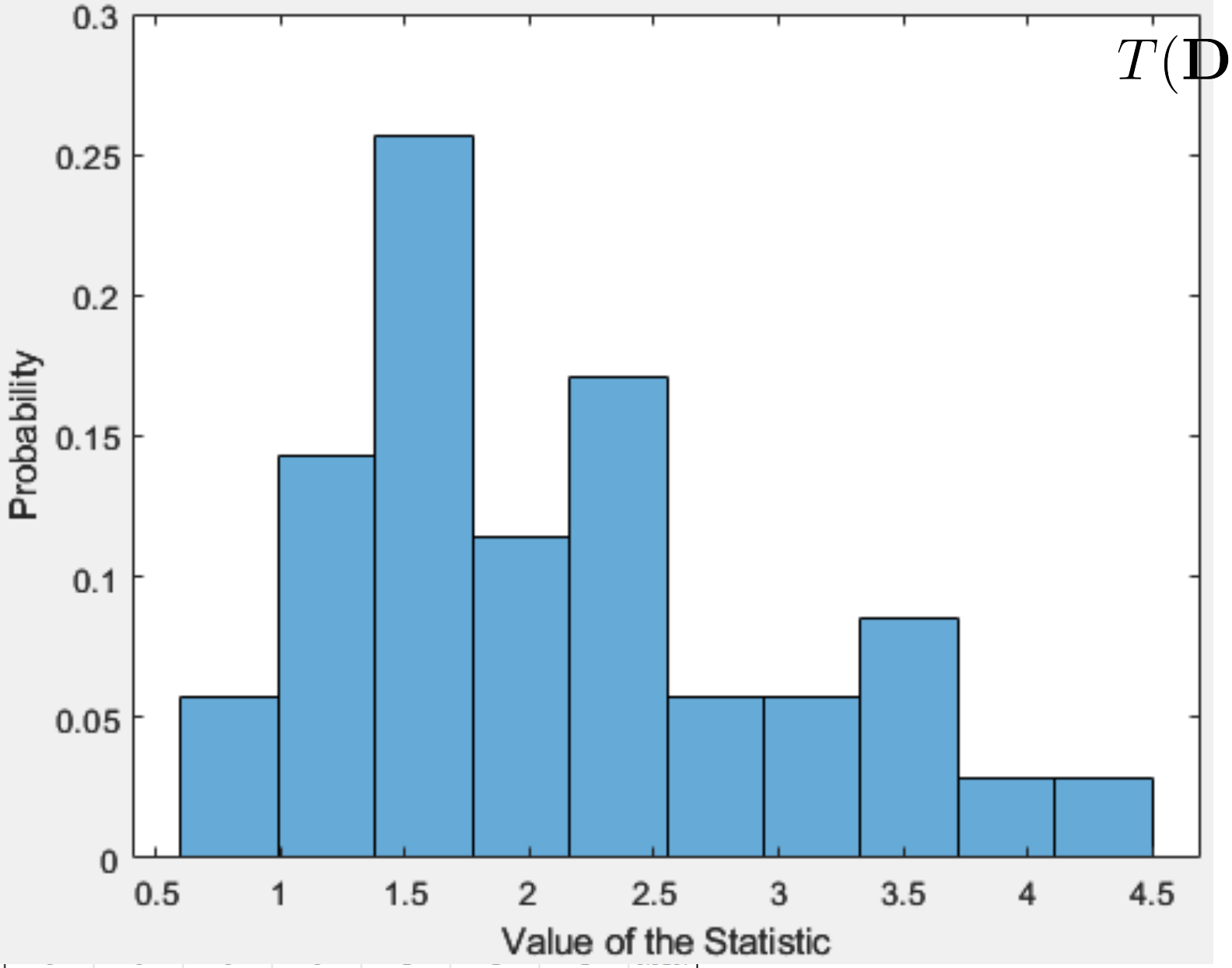
Fisher's Exact Test

Serial #	D1	D2	D3	D4	D5	D6	D7	T
1	1	1	1	0	0	0	0	3.7037
2	1	1	0	1	0	0	0	2.3188
3	1	1	0	0	1	0	0	2.1111
4	1	1	0	0	0	1	0	1.4286
5	1	1	0	0	0	0	1	1.5802
6	1	0	1	1	0	0	0	3.3684
7	1	0	1	0	1	0	0	3.0667
8	1	0	1	0	0	1	0	2.1111
9	1	0	1	0	0	0	1	2.3188
10	1	0	0	1	1	0	0	1.92
11	1	0	0	1	0	1	0	1.2874
12	1	0	0	0	1	1	0	1.1556
13	1	0	0	0	1	0	1	1.2874
14	1	0	0	0	0	1	1	0.8081
15	0	1	1	1	0	0	0	4.5
16	0	1	1	0	1	0	0	4.0784
17	0	1	1	0	0	1	0	2.7937
18	0	1	1	0	0	0	1	3.0667
19	0	1	0	1	1	0	0	2.5455
20	0	1	0	1	0	1	0	1.7436
21	0	1	0	1	0	0	1	1.92
22	0	1	0	0	1	1	0	1.5802
23	0	1	0	0	1	0	1	1.7436
24	0	1	0	0	0	1	1	1.1556
25	0	0	1	1	1	0	0	3.7037
26	0	0	1	1	0	1	0	2.5455
27	0	0	1	1	0	0	1	2.7937
28	0	0	1	0	1	1	0	2.3188
29	0	0	1	0	1	0	1	2.5455
30	0	0	1	0	0	1	1	1.7436
31	0	0	0	1	1	1	0	1.4286
32	0	0	0	1	1	0	1	1.5802
33	0	0	0	1	0	1	1	1.0323
34	0	0	0	0	1	1	1	0.9167
35	1	0	0	1	0	0	1	1.4286

$$T(\mathbf{D}, \mathbf{Y}^{obs}) = \bar{Y}^{obs}(1)/\bar{Y}^{obs}(0)$$

Fisher's Exact Test

Serial #							
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
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30							
31							
32							
33							
34							
35	1	0	0	1	0	0	1	1.4286

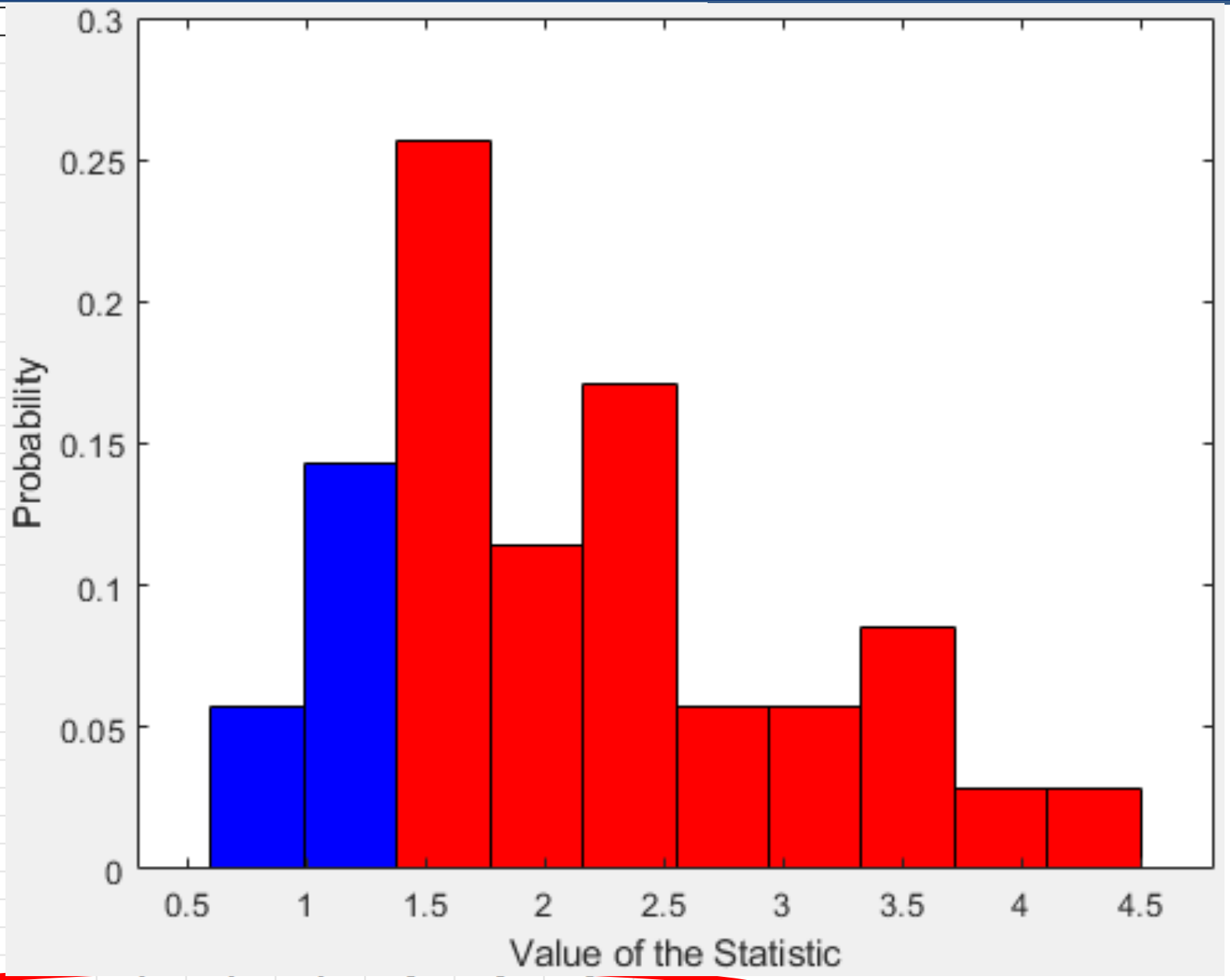


$$T(\mathbf{D}, \mathbf{Y}^{obs}) = \bar{Y}^{obs}(1)/\bar{Y}^{obs}(0)$$

The distribution of the statistic relies on 2 things: (1) the Fisher sharp null hypothesis; (2) the specific randomization procedure

Fisher's Exact Test

Serial #								
1								
2								
3								
4								
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9								
10								
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12								
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20								
21								
22								
23								
24								
25								
26								
27								
28								
29								
30								
31								
32								
33								
34								
35	1	0	0	1	0	0	1	1.4286



In this experiment, the chance of observing the statistic being 1.4 or even more extreme (i.e., greater than 1.4) is 80%.

Fisher's Exact Test

The hypothesis must allow us to fully infer the missing value (the potential outcome that is unobservable).

Select a test statistic (a function of the randomized assignment and the observed outcome). Also specify what counts as a more extreme result.

Derive the probability distribution of the test statistic under the null hypothesis and the randomization procedure

Check the probability of observing the specific value (and even more extreme value) of the test statistic in your specific experiment

Randomized Assignment Procedure

1. Bernoulli Trials
2. Completely Randomized Experiment
3. Stratified Randomized Experiment (Randomized Blocks)
4. Paired Randomized Experiment (A special case of 3)

The key difference among these 4 types of randomization procedure is in the set of assignment vectors with positive probability

$$D_i = \begin{cases} 1 & \text{if unit } i \text{ received the treatment,} \\ 0, & \text{otherwise} \end{cases} \qquad \mathbf{D} \triangleq (D_1, D_2, ..., D_N) \in \{0,1\}^N$$

$\mathbb{D} = \{\mathbf{D}\} = \{0,1\}^N$ is the assignment outcome of all N subjects

Randomized Assignment Procedure

What are the possible value of this vector $\mathbf{D}$?

$\mathbf{D} = (0, 0, 1, 1, 0, 1, 0, ..., 1)$

$\mathbf{D} = (0, 0, 0, 0, 0, 0, 0, ..., 0)$ All entries are 0

Independent coin flips can produce this assignment.

But is it helpful for comparing the two groups?

$\mathbf{D} = (1, 1, 1, 1, 1, 1, 1, ..., 1)$ All entries are 1

In the space of $\{0,1\}^N$, there are some assignment outcomes less useful

Some randomized designs rule out empty groups or other undesirable allocations

$\mathbb{D}^+$ is the set of values of $\{0,1\}^N$ that have strictly positive probabilities

Simplest version: for each subject, you independently flip a fair coin

$\mathbb{D}^+ = \{0,1\}^N$ Any realizations of the binary assignment vector can happen

$\mathbf{D} = (0,0,1,1,0,1,0,...,1)$ occurs with probability 0.5^N

$\mathbf{D} = (0,0,0,0,0,0,0,...,0)$ occurs with probability 0.5^N

$\mathbf{D} = (1,1,1,1,1,1,1,...,1)$ occurs with probability 0.5^N

Some variations of Bernoulli Trial

1. The coin does not have to be a fair coin
2. You can actually flip a different coin for different subject in order to do assignment as long as the coin is independent of the potential outcome of the subject

$\mathbb{D}^+ = \{0,1\}^N$ Any realizations of the binary assignment vector can happen

Bernoulli Trial is easy to implement it is useful especially when you do not know how many subjects will join the experiment in advance.

Online A/B testing: which version of the webpage customers like more?



Online A/B testing: which version leads to more completed donations?

If you anticipate many subjects to participate in your experiment but you do not know in advance about how many will participate, Bernoulli Trial is a good option

Randomized Assignment Procedure: Completely Randomized Experiment

If you know how many subjects will show up in your experiment but the total number is small, then Bernoulli Trial may not be a good option.

In the Bernoulli Trial, the number of subjects being assigned to one group is a random variable.

In the Completely Randomized Experiment, the number of subjects being assigned to one group is a pre-determined fixed number

e.g., you know there will be 10 subjects attend your experiment, you just random choose 5 to be assigned to the treatment group.

There are $\binom{10}{5} = 252$ possibilities, e.g., (1,3,7,9,10), (2,4,5,8,9),…
You can let computer generate all of these 252 cases and choose one with probability 1/252

Randomized Assignment Procedure: Completely Randomized Experiment

$$\mathbb{P}\{\mathbf{D}\} = \begin{cases} \binom{N}{N_T}^{-1}, & \text{if } \sum_{i=1}^{N} D_i = N_T, \\ 0, & \text{otherwise.} \end{cases} \qquad \mathbb{D}^+ = \left\{ \mathbf{D} \in \mathbb{D} \middle| \sum_{i=1}^{N} D_i = N_T \right\}$$

The Completely Randomized Experiment excludes some of the assignment vector, e.g., 10 subjects in total, you will never see an assignment such that all 10 subjects are assigned to one group, which could happen under a Bernoulli Trial

The Completely Randomized Experiment is more efficient and is widely used in the lab experiment when there are not many subjects in total and the total number is known in advance. But it is not very useful in the online A/B test setting

Randomized Assignment Procedure: Completely Randomized Experiment

When there are important covariates observable a priori and possibly highly associated with the potential outcome, the Completely Randomized design has some other drawbacks.

e.g., 12 subjects in total with 6 men and 6 women. The task is to test the treatment effect of a new drug on blood pressure reduction.

Gender affects blood pressure substantially. Although the Completely Randomized design can ensure that there are always 6 people in the control and 6 people in the treatment group. However, it cannot prevent a case like all 6 men being assigned to one group from happening.

If that is the case, then any difference in performance metric between 2 groups cannot simply be attributed to the treatment alone, as the gender difference may also be able to explain that

This implies that we may also want to exclude more useless assignment vector

Randomized Assignment Procedure : Stratified Randomized Experiment

When to apply Stratified Randomized Experiment (or Block Design)?

If you can observe some covariates (attributes or characteristics) of the subjects in advance

Common covariates are: gender, age, height, weight, etc.

The goal of the block design is to keep the treatment and control groups approximately balanced on the covariates used to define the blocks.

e.g., prevent one group has most elder subjects and the other group has most young subjects

	M	F
Y	10	6
O	9	4

Here we have 4 blocks and do a completely randomized design within each block

	M	F
Y	(5,5)	(3,3)
O	(4,5)	(2,2)

	Control	Treatment
M	9	10
F	5	5
Y	8	8
O	6	7

Randomization Procedure: Paired Randomization

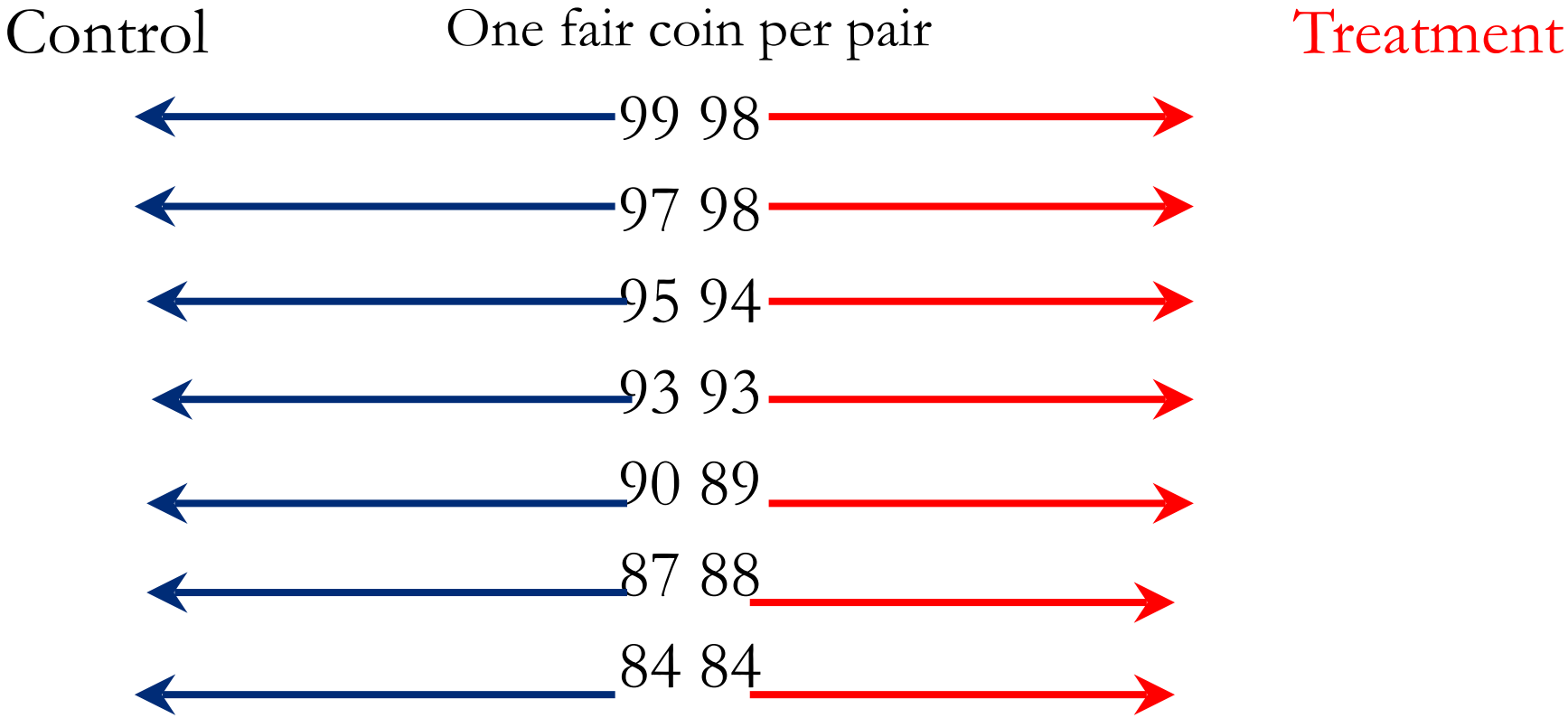
Basic Idea: If you have data on pre-treatment characteristic X, then why leave it to "pure chance" to balance them?

"Block what you can and randomize what you cannot!"

There is an extreme version of the block design, the paired randomization: Each block only contains 2 subjects and a fair coin is tossed to decide which subject goes to the control group and which subject goes to the treatment group.

Randomization Procedure: Paired Randomization

Education experiment with one covariate: pre-test score



Block design and paired randomization restrict assignments to improve covariate balance. They can improve precision, but they require more preparation before the experiment.

Random Sample From A Super-Population

So far, our focus is only the set of subjects who participate in our experiment

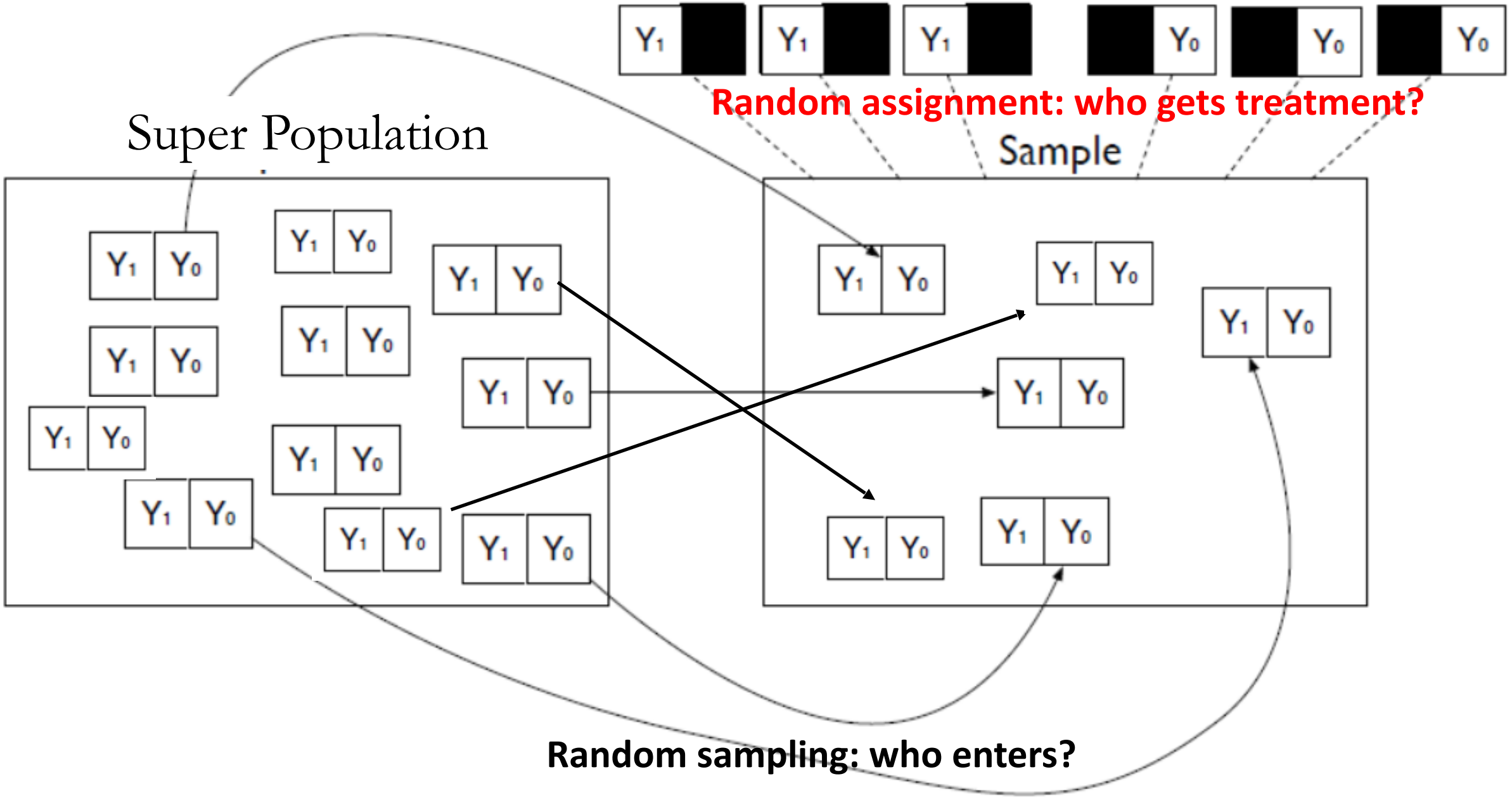
These subjects are our target population

In practice, our purpose of conducting experiment is to say something about a more general population and hence we view the set of subjects who participate in our experiment as a ***sample*** from the ***super-population***

This sampling process introduces an additional layer of uncertainty in causal inference:

1. Assume N units are randomly sampled from the "super population"
2. N_T units (from the N units) are then randomly assigned to the treatment group

Random Sample From A Super-Population



Random Sample From A Super-Population

How does this point of view affect our causal inference, in terms of:

- Point estimates: is observed difference in means still unbiased?
- Uncertainty estimates: how can we incorporate both sources of variation?

$$\tau_{fs} = \frac{1}{N}\sum_{i=1}^{N} Y_i(1) - \frac{1}{N}\sum_{i=1}^{N} Y_i(0)$$

Random Sampling Uncertainty

$$\tau_{sp} = \mathbb{E}_{sp}[Y_i(1)] - \mathbb{E}_{sp}[Y_i(0)]$$

Unbiased

Uncertainty comes from randomized assignment

Unbiased?

$$\hat{\tau}_{fs} = \bar{Y}_T^{obs} - \bar{Y}_C^{obs} = \frac{1}{N_T}\sum_{i\in T} Y_i(1) - \frac{1}{N_C}\sum_{i\in C} Y_i(0)$$

Will this extra layer of uncertainty makes the things worse?

Assume completely randomized assignment

Random Sample From A Super-Population

In practice, it is often very difficult to conclude that the set of subjects in your experiment are random draw from the super population you want to target. In most experiment, subjects choose to join voluntarily and are not drawn randomly by the experimenter.

$$\tau_{fs} = \frac{1}{N}\sum_{i=1}^{N} Y_i(1) - \frac{1}{N}\sum_{i=1}^{N} Y_i(0)$$

Random Sampling Uncertainty

$$\tau_{sp} = \mathbb{E}_{sp}[Y_i(1)] - \mathbb{E}_{sp}[Y_i(0)]$$

Unbiased

Uncertainty comes from randomized assignment

Unbiased?

$$\hat{\tau}_{fs} = \bar{Y}_T^{obs} - \bar{Y}_C^{obs} = \frac{1}{N_T}\sum_{i\in T} Y_i(1) - \frac{1}{N_C}\sum_{i\in C} Y_i(0)$$

It is unbiased for τ_{sp} as long as the samples are representative of the population

Observed Difference in Average Between Two Groups Revisited

$\bar{Y}_T^{obs} = \frac{1}{N_T} \sum_{i \in T} Y_i(1)$ is an unbiased estimator of $\mathbb{E}[Y_i^{obs} | D_i = 1]$

$\bar{Y}_C^{obs} = \frac{1}{N_C} \sum_{i \in C} Y_i(0)$ is an unbiased estimator of $\mathbb{E}[Y_i^{obs} | D_i = 0]$

$\bar{Y}_T^{obs} - \bar{Y}_C^{obs}$ is an unbiased estimator of

$\underbrace{\mathbb{E}[Y_i^{obs} | D_i = 1] - \mathbb{E}[Y_i^{obs} | D_i = 0]}$

Expected observed difference in the super population

$= \mathbb{E}[Y_i(1) | D_i = 1] - \mathbb{E}[Y_i(0) | D_i = 0]$

$= \mathbb{E}[Y_i(1) | D_i = 1] - \mathbb{E}[Y_i(0) | D_i = 1]$

$+ \mathbb{E}[Y_i(0) | D_i = 1] - \mathbb{E}[Y_i(0) | D_i = 0]$

Recall: $Y_i^{obs} = D_i Y_i(1) + (1 - D_i) Y_i(0)$

Observed Difference in Average Between Two Groups Revisited

$$\underbrace{\mathbb{E}[Y_i^{obs}|D_i=1]-\mathbb{E}[Y_i^{obs}|D_i=0]}_{\text{Expected observed difference in the super population}}$$

$$= \mathbb{E}[Y_i(1)|D_i=1]-\mathbb{E}[Y_i(0)|D_i=0]$$

Factual

$$= \underbrace{\mathbb{E}[Y_i(1)|D_i=1]-\mathbb{E}[Y_i(0)|D_i=1]}_{\text{Average treatment effect on the treated}}$$

Counterfactual

$$+\underbrace{\mathbb{E}[Y_i(0)|D_i=1]-\mathbb{E}[Y_i(0)|D_i=0]}_{\textbf{Selection bias}}$$

In general, the selection bias $\neq 0$

The observed difference alone cannot generally be interpreted as a causal effect.

When the assignment is independent of the potential outcome, then the selection bias is 0. Randomized assignment ensures that

$$\mathbb{E}[Y_i(0)|D_i=1]-\mathbb{E}[Y_i(0)|D_i=0]=\mathbb{E}[Y_i(0)]-\mathbb{E}[Y_i(0)]=0$$

Observed Difference in Average Between Two Groups Revisited

$$\bar{Y}_T^{obs} = \frac{1}{N_T}\sum_{i\in T} Y_i(1) \quad \bar{Y}_C^{obs} = \frac{1}{N_C}\sum_{i\in C} Y_i(0)$$

$$\mathbb{E}[Y_i^{obs}|D_i=1] - \mathbb{E}[Y_i^{obs}|D_i=0]$$

Expected observed difference in the super population

Factual **Counterfactual**

$$= \mathbb{E}[Y_i(1)|D_i=1] - \mathbb{E}[Y_i(0)|D_i=1]$$

Average treatment effect on the treated

$$= \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)]$$

Since D_i is independent of $Y_i(1)$ and $Y_i(0)$

$$= \tau_{sp}$$

Random sampling ensures that the subjects who participate in your experiment are representatives of the super population

Randomized assignment establishes the validity of using the observed difference in average performance ($\hat{\tau}_{fs}$) as an unbiased estimator of the super population average causal effect (τ_{sp})